others, developed models that explain how a supermassive black hole can energize radio galaxies, quasars, and active galactic nuclei (Chapter 9).

Sakharov, Andrei Dmitrievich (1921–1989). Soviet theoretical physicist; invented key ideas that underlie the Soviet hydrogen bomb (Chapter 6); close friend, associate, and competitor of Zel'dovich's (Chapters 6 and 7); later became the leading Soviet dissident and, after glasnost, Soviet saint.

Schwarzschild, Karl (1876–1916). German astrophysicist; discovered the Schwarzschild solution of the Einstein field equation, which describes the spacetime geometry of a nonspinning star that is either static or imploding, and also describes a nonspinning black hole (Chapter 3); discovered the solution of the Einstein equation for the interior of a constant-density star—a solution that Einstein used to argue that black holes cannot exist (Chapter 3).

Sciama, Dennis (b. 1926). British astrophysicist and mentor for British researchers on black holes (Chapters 7 and 13).

Tenkolsky, **Sanl** A. (b. 1947). South African born, American theoretical physicist; student of Thorne's; invented and developed the formalism by which perturbations of spinning black holes are analyzed and, with Press, used his formalism to show that black holes are stable against small perturbations (Chapters 7 and 12); with Shapiro, discovered evidence that the laws of physics might permit naked singularities to form in our Universe (Chapter 13).

Thorne, **Kip** S. (b. 1940). American theoretical physicist; student of Wheeler's; proposed the hoop conjecture which describes when black holes can form in an imploding star, and developed evidence for it (Chapter 7); made estimates of the gravitational waves from astrophysical sources and contributed to ideas and plans for the detection of those waves (Chapter 10); with others, developed the membrane paradigm for black holes (Chapter 11); developed ideas about the statistical origin of the entropy of a black hole (Chapter 12); probed the laws of physics by means of thought experiments about wormholes and time machines (Chapter 14).

Wald, Robert M. (b. 1947). American theoretical physicist; student of Wheeler's; contributed to the Teukolsky formalism for analyzing perturbations of black holes and its applications (Chapter 7); with others, developed an understanding of how electric fields behave outside a black hole—an understanding that underlies the membrane paradigm (Chapter